Q1: 16 March (Shift 1) - Single Correct

Let a complex number z, $|z| \neq 1$,
satisfy $\log_{\frac{1}{\sqrt{2}}}\left(\frac{|z|+11}{(|z|-1)^2}\right) \leq 2$. Then, the largest value of $|z|$ is equal to

(1) 8

(2) 7

(3) 6

(4) 5

Q2: 16 March (Shift 1) - Numerical

Let z and w be two complex numbers such that

$$\mathrm{w} = \mathrm{z}\bar{\mathrm{z}} - 2\mathrm{z} + 2, \left|\frac{\mathrm{z}+\mathrm{i}}{\mathrm{z}-3\mathrm{i}}\right| = 1 \quad \text{and} \quad \operatorname{Re}(\mathrm{w}) \text{ has}$$

minimum value. Then,the minimum value of

$\mathrm{n} \in \mathbb{N}$ for which w^{n} is real, is equal to________

Q3: 16 March (Shift 2) - Single Correct

The least value of $|z|$ where z is complex number which satisfies the inequality

$$\exp\left(\frac{(|z|+3)(|z|-1)}{||z|+1|}\log_e 2\right) \geq \log_{\sqrt{2}}|5\sqrt{7}+9i|$$

$i = \sqrt{-1}$, is equal to :

(1) 3

(2) $\sqrt{5}$

(3) 2

(4) 8

Q4: 17 March (Shift 1) - Single Correct

The area of the triangle with vertices $\mathrm{A(z), B(iz)}$ and $\mathrm{C(z+iz)}$ is :

(1) 1

(2) $\frac{1}{2}|\mathrm{z}|^2$

(3) $\frac{1}{2}$

(4) $\frac{1}{2}|\mathrm{z}+\mathrm{iz}|^2$

Q5: 17 March (Shift 2) - Single Correct

Let S_1, S_2 and S_3 be three sets defined as

$\mathrm{S}_1=\{\mathrm{z}\in\mathbb{C}:|\mathrm{z}-1|\leq\sqrt{2}\}$

$\mathrm{S}_2=\{\mathrm{z}\in\mathbb{C}:\mathrm{Re}((1-\mathrm{i})\mathrm{z})\geq 1\}$

$\mathrm{S}_3=\{\mathrm{z}\in\mathbb{C}:\mathrm{Im}(\mathrm{z})\leq 1\}$

Then the set $\mathrm{S}_1\cap\mathrm{S}_2\cap\mathrm{S}_3$

(1) is a singleton

(2) has exactly two elements

(3) has infinitely many elements

(4) has exactly three elements

Q6: 18 March (Shift 1) - Single Correct

If the equation $\mathrm{a}|\mathrm{z}|^2+\overline{\bar{\alpha}\mathrm{z}+\alpha\bar{\mathrm{z}}}+\mathrm{d}=0$ represents a circle where $\mathrm{a, d}$ are real constants then which of the following condition is correct?

(1) $|\alpha|^2-\mathrm{ad}\neq 0$

(2) $|\alpha|^2-ad>0$ and $a\in R-\{0\}$

(3) $|\alpha|^2-ad\geq 0$ and $a\in R$

(4) $\alpha=0, \mathrm{a, d}\in\mathrm{R}^+$

Q7: 18 March (Shift 1) - Numerical

Let z_1, z_2 be the roots of the equation $z^2 + az + 12 = 0$ and z_1, z_2 form an equilateral triangle with origin. Then, the value of lal is

Q8: 18 March (Shift 2) - Single Correct

Let a complex number be $w = 1 - \sqrt{3}i$. Let another complex number z be such that $|zw| = 1$ and $\arg(z) - \arg(w) = \frac{\pi}{2}$. Then the area of the triangle with vertices origin, z and w is equal to:

(1) 4

(2) $\frac{1}{2}$

(3) $\frac{1}{4}$

(4) 2

Answer Key

Q1 (2)	**Q2** (4)	**Q3** (1)	**Q4** (3)
Q5 (3)	**Q6** (2)	**Q7** (6)	**Q8** (2)

Q1 (2)

$\log_{\frac{1}{\sqrt{2}}}\left(\frac{|z|+11}{(|z|-1)^2}\right) \leq 2$

$\frac{|z|+11}{(|z|-1)^2} \geq \frac{1}{2}$

$2|z| + 22 \geq (|z|-1)^2$

$2|z| + 22 \geq |z|^2 + 1 - 2|z|$

$|z|^2 - 4|z| - 21 \leq 0$

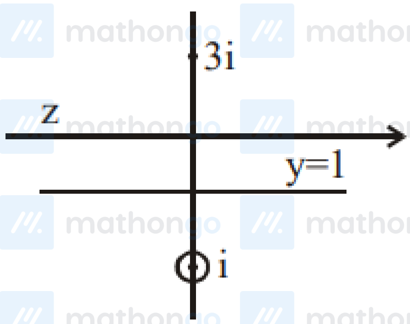
$\Rightarrow |z| \leq 7$

$\therefore$ Largest value of $|z|$ is 7

Q2 (4)

$$\begin{aligned}
&\omega = z\bar{z} - 2z + 2\\
&\left|\frac{z+i}{z-3i}\right| = 1\\
&\Rightarrow \quad |z+i| = |z-3i|\\
&\Rightarrow \quad z = x + i, \quad x \in \mathbb{R}\\
&\omega = (x+i)(x-i) - 2(x+i) + 2\\
&\quad = x^2 + 1 - 2x - 2i + 2\\
&\operatorname{Re}(\omega) = x^2 - 2x + 3\\
&\text{For } \min(\operatorname{Re}(\omega)), x = 1\\
&\Rightarrow \quad \omega = 2 - 2i = 2(1-i) = 2\sqrt{2}e^{-i\frac{\pi}{4}}\\
&\omega^n = (2\sqrt{2})^n e^{-i\frac{n\pi}{4}}
\end{aligned}$$

For real \& minimum value of n n $= 4$



Q3 (1)

$\exp\left(\frac{(|z|+3)(|z|-1)}{||z|+1|}\ln 2\right) \geq \log_{\sqrt{2}}|5\sqrt{7}+9i|$

$\Rightarrow \quad 2^{\frac{(|z|+3)(|z|-1)}{(|z|+1)}} \geq \log_{\sqrt{2}}(16)$

$\Rightarrow \quad 2^{\frac{(|z|+3)(|z|-1)}{(|z+1)}} \geq 2^3$

$\Rightarrow \quad \frac{(|z|+3)(|z|-1)}{(|z|+1)} \geq 3$

$\Rightarrow \quad (|z|+3)(|z|-1) \geq 3(|z|+1)$

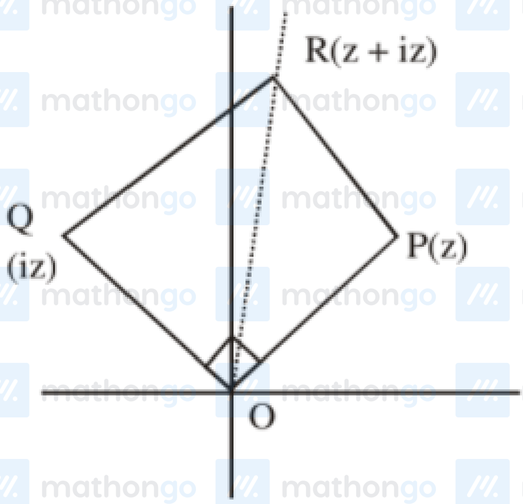
$|z|^2 + 2|z| - 3 \geq 3|z| + 3$

$\Rightarrow |z|^2 + |z| - 6 \geq 0$

$\Rightarrow (|z|-3)(|z|+2) \geq 0 \Rightarrow |z|-3 \geq 0$

$\Rightarrow |z| \geq 3 \quad \Rightarrow |z|_{\min} = 3$

Q4 (3)



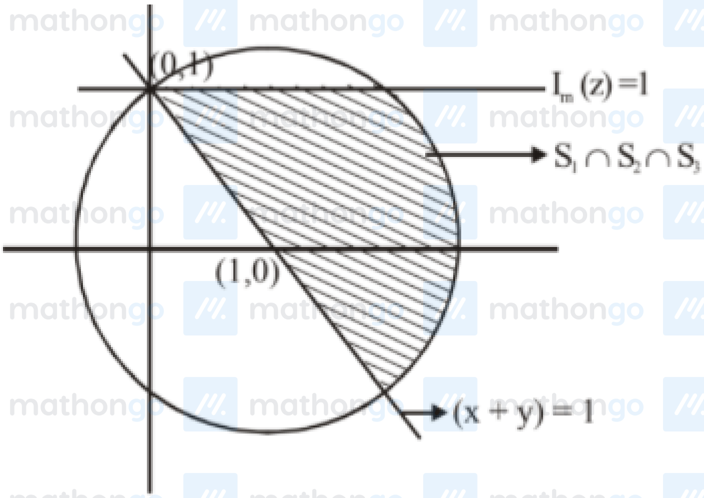
$A = \frac{1}{2}|z||iz|$

$= \frac{|z|^2}{2}$

Q5 (3)

For $|z-1| \leq \sqrt{2}$, z lies on and inside the circle

of radius $\sqrt{2}$ units and centre (1,0) .



For S_2 Let $z = x +$ iy Now, $(1 - i)(z) = (1 - i)(x + iy)$

$\operatorname{Re}((1 - i)z) = x + y$

$\Rightarrow x + y \geq 1$

$\Rightarrow S_1 \cap S_2 \cap S_3$ has infinity many elements

Q6 (2)

$\mathrm{az\bar{z}} + \alpha\bar{\mathrm{z}} + \bar{\alpha}\mathrm{z} + \mathrm{d} = 0 \rightarrow$ Circle

centre $= \frac{-\alpha}{\mathrm{a}} \quad 2 = \sqrt{\frac{\alpha\bar{\alpha}}{\mathrm{a}^2} - \frac{\mathrm{d}}{\mathrm{a}}} = \sqrt{\frac{\alpha\bar{\alpha} - \mathrm{ad}}{\mathrm{a}^2}}$

So $|\alpha|^2 - \mathrm{ad} > 0 \& \mathrm{a} \in \mathrm{R} - \{0\}$

Q7 (6)

If $0, \mathrm{z}, \mathrm{z}_2$ are vertices of equilateral triangles

$\Rightarrow a^2 + z_1^2 + z_2^2 = 0\,(z_1 + z_2) + z_1 z_2$

$\Rightarrow (z_1 + z_2)^2 = 3 z_1 z_2$

$\Rightarrow \mathrm{a}^2 = 3 \times 12$

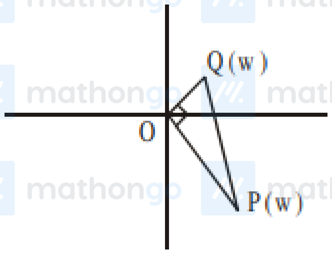
$\Rightarrow |\mathrm{a}| = 6$

Q8 (2)

$\mathrm{w}=1-\sqrt{3}. i \Rightarrow|\mathrm{wl}=2$

Now, $|\mathrm{z}|=\frac{1}{|\mathrm{w}|} \Rightarrow|\mathrm{z}|=\frac{1}{2}$

and $\operatorname{amp}(\mathrm{z})=\frac{\pi}{2}+\operatorname{amp}(\mathrm{w})$



$\Rightarrow$ Area of triangle $=\frac{1}{2} \cdot \mathrm{OP}. \mathrm{OQ}$

$=\frac{1}{2} \cdot 2 \cdot \frac{1}{2}=\frac{1}{2}$